# Business Question and Visualization Report

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| Date | 24 September 2024 |
| Team ID | [Add Team ID] |
| Project Name | Global Fuel Prices & Taxation Analysis Dashboard |
| Maximum Marks | 5 Marks |

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

## Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

**1. Which countries tax fuel the most, and how does tax vary across the world?**

**Visualization:** Filled map of average fuel tax by country (Sheet 1).

**Insight:** Rwanda, New Zealand, Germany, Turkey and Denmark have the highest average tax at about 39%. Iran, Morocco and Qatar are lowest at about 14%.

*[Insert Tableau screenshot of Sheet 1]*

**2. How do tax rates and subsidy levels compare across regions?**

**Visualization:** Region by subsidy level heat table with average tax (Sheet 2).

**Insight:** Oceania (about 38.7%) and Europe (about 34.6%) tax fuel most; South America (about 18.2%) and the Middle East (about 19.6%) tax least.

*[Insert Tableau screenshot of Sheet 2]*

**3. Do higher subsidies lead to lower fuel prices?**

**Visualization:** Bar chart of average fuel price by subsidy level (Sheet 3).

**Insight:** Average price falls from about $3.09 per litre at Low subsidy to about $0.13 at Very High.

*[Insert Tableau screenshot of Sheet 3]*

**4. How are diesel prices trending and where are they heading?**

**Visualization:** Line chart of monthly diesel price by region with a forecast (Sheet 4).

**Insight:** Average diesel rose from about $1.39 per litre in 2020 to about $2.47 in 2026, and Europe and Oceania stay highest.

*[Insert Tableau screenshot of Sheet 4]*

**5. Where is diesel cheapest and costliest?**

**Visualization:** Ranked bar charts of the top 10 and bottom 10 countries by diesel price (Sheet 5).

**Insight:** Hong Kong ($5.10), Ireland ($4.55) and Norway ($4.43) are costliest; Venezuela, Iran and Libya are below $0.10.

*[Insert Tableau screenshot of Sheet 5]*

**6. How risky and volatile are diesel prices by region and year?**

**Visualization:** Heat map of diesel volatility by region and year (Sheet 6).

**Insight:** The Middle East is the most volatile (about 1.18 to 1.22) and Oceania the least (about 0.06 to 0.20).

*[Insert Tableau screenshot of Sheet 6]*

**7. Do wealthier countries tax fuel more?**

**Visualization:** Bar chart of average tax by income level (Sheet 7).

**Insight:** High-income countries tax fuel at about 34.7%, against about 22.9% for Low and 21.1% for Middle.

*[Insert Tableau screenshot of Sheet 7]*

**8. How are subsidy levels spread across income groups?**

**Visualization:** 100% stacked bar chart of subsidy level by income group (Sheet 8).

**Insight:** About 87.5% of High-income records are at the Low subsidy level, so subsidies are concentrated in Low and Middle income countries.

*[Insert Tableau screenshot of Sheet 8]*

**9. How do petrol, diesel and LPG prices differ by income level?**

**Visualization:** Highlight table of average price by fuel type and income level (Sheet 9).

**Insight:** High-income countries pay about $3.55 for petrol and $3.35 for diesel, against about $1.39 and $1.28 in Low-income countries.

*[Insert Tableau screenshot of Sheet 9]*